

$$27- \int_{\pi/3}^{\pi/2} \frac{\sin^2 n}{\sqrt{1-\cos n}} dn$$

Sol:-

$$\int \frac{\sin^2 n}{\sqrt{1-\cos n}} \times \frac{\sqrt{1+\cos n}}{\sqrt{1+\cos n}} dn$$

$$\int \frac{\sin^2 n \times \sqrt{1+\cos n}}{\sqrt{1-\cos^2 n}} dn \Rightarrow \int \frac{\sin^2 n \times \sqrt{1+\cos n}}{\sqrt{\sin^2 n}} dn$$

$$\int \frac{\sin^2 n \times \sqrt{1+\cos n}}{\sin n} dn \Rightarrow \int (1+\cos n)^{1/2} \sin n dn$$

Let $u = 1 + \cos n$ $du = -\sin n dn$
 $n = \pi/3$ $u = 1 + \cos(\pi/3) = 3/2$
 $n = \pi/2$ $u = 1 + \cos(\pi/2) = 1$

$$\int_{3/2}^{1} (1+\cos n)^{1/2} \sin n dn = \int_{3/2}^1 \sqrt{u} du \Rightarrow \int_1^{3/2} (u)^{1/2} du \Rightarrow \frac{u^{1/2+1}}{1/2+1} + C$$

$$\left[\frac{u^{3/2}}{3/2} + C \right]_1^{3/2} \Rightarrow \frac{2}{3} u^{3/2} \Big|_1^{3/2}$$

put Limit upper Limit
 $u = 3/2$ $3/2$ ~~Lower~~ Limit

$$\Rightarrow \frac{2}{3} \left(\frac{3}{2} \right)^{3/2} = \frac{\sqrt{6}}{2}$$

Lower ~~upper~~ Limit upper - Lower

$$\Rightarrow \frac{2}{3} (1)^{3/2} = \frac{2}{3} \quad \frac{\sqrt{6}}{2} - \frac{2}{3}$$



$$28) \int_0^{\pi/6} \sqrt{1 + \sin n} \, dn$$

Sol:-

$$\int \sqrt{1 + \sin n} \times \frac{\sqrt{1 - \sin n}}{\sqrt{1 - \sin n}} \, dn$$

$$\int \frac{\sqrt{1 - \sin^2 n}}{\sqrt{1 - \sin n}} \, dn \Rightarrow \int \frac{\sqrt{\cos^2 n}}{\sqrt{1 - \sin n}} \, dn$$

$$\int \frac{\cos n}{\sqrt{1 - \sin n}} \, dn$$

$$u = 1 - \sin n \Rightarrow du = -\cos n \, dn$$

$$n = \pi/6 \quad u = 1 - \sin(\pi/6) = 1/2$$

$$n = 0 \quad u = 1 - \sin(0) = 1$$

$$- \int \frac{du}{\sqrt{u}}$$

$$\Rightarrow - \int_0^{\pi/6} u^{-1/2} \, du$$

$$\Rightarrow - \frac{u^{-1/2+1}}{1/2+1} + c \Bigg]_{1/2}^1$$

$$- \frac{u^{1/2}}{1/2} + c \Bigg]_{1/2}^1$$

$$\Rightarrow - \frac{2}{1} \sqrt{u} \Bigg]_{1/2}^1$$

$$u = 1/2 \quad \text{upper limit} \quad \text{Lower}$$

$$- \frac{2}{1} \cdot \left(\frac{1}{2}\right)^{1/2} \Rightarrow \frac{\sqrt{2}}{2} \times -\frac{2}{1} = -\sqrt{2}$$

$$u = 1 \quad \text{lower} \quad \text{upper}$$

$$- \frac{2}{1} \cdot (1)^{1/2} \Rightarrow -2$$

$$\text{upper} - \text{Lower}$$

$$-2 - (-\sqrt{2})$$

$$= -2 + \sqrt{2}$$

$$+ 2 - \sqrt{2} \Rightarrow 2 - \sqrt{2}$$



$$29) \int_{\pi/6}^{\pi} \frac{\cos^4 n}{\sqrt{1-\sin n}} dn$$

Sol:-

$$\int \frac{\cos^4 n}{\sqrt{1-\sin n}} \times \frac{\sqrt{1+\sin n}}{\sqrt{1+\sin n}} dn$$

$$\int \frac{\cos^4 n \sqrt{1+\sin n}}{\sqrt{1-\sin^2 n}} dn \Rightarrow \int \frac{\cos^4 n \sqrt{1+\sin n}}{-\cos^2 n} dn$$

$$-\int \cos^3 n \sqrt{1+\sin n} dn$$

$$-\int \cos^2 n \cdot \cos n \sqrt{1+\sin n} dn$$

$$-\int (1-\sin^2 n) \cos n \sqrt{1+\sin n} dn$$

$$-\int (\cos n + \cos n \sin^2 n) \sqrt{1+\sin n} dn$$

$$-\int \cos n \sqrt{1+\sin n} dn + \int \cos n \sin^2 n \sqrt{1+\sin n} dn$$

$$u = 1 + \sin n \quad du = \cos n dn = \sin n = u - 1$$

$$-\int \sqrt{1+\sin n} \cos n dn + \int \sin^2 n \sqrt{1+\sin n} \cos n dn$$

$$-\int \sqrt{u} du + \int (u-1)^2 \sqrt{u} du$$

$$-\int u^{1/2} du + \int (u^2 - 2u + 1) u^{1/2} du$$

$$-\int u^{1/2} du + \int u^{2+1/2} - 2u^{1+1/2} + u^{1/2} du$$



$$\int -\int u^{1/2} du + \int u^{5/2} - 2u^{3/2} + u^{1/2} du$$

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$$-\frac{u^{1/2+1}}{1/2+1} + \frac{u^{5/2+1}}{5/2+1} - \frac{2u^{3/2+1}}{3/2+1} + \frac{u^{1/2+1}}{1/2+1} + C$$

$$-\frac{u^{3/2}}{3/2} + \frac{u^{7/2}}{7/2} - \frac{2u^{5/2}}{5/2} + \frac{u^{3/2}}{3/2} + C$$

$$\cancel{\frac{2}{3} \times u^{3/2}} + \frac{2}{7} u^{7/2} - \cancel{\frac{2 \times 2}{5} u^{5/2}} + \cancel{\frac{2}{3} u^{3/2}} + C$$

$$\frac{2}{7} u^{7/2} - \frac{4}{5} u^{5/2} + C$$

$$u = 1 + 8 \sin n$$

$$n = \pi$$

$$u = 1$$

→ upper

$$u = 1 + 8 \sin n$$

$$n = 5\pi/6$$

$$u = 3/2$$

→ Lower

$$\left[\frac{2}{7} u^{7/2} - \frac{4}{5} u^{5/2} + C \right]_1^{3/2}$$

$$u = 3/2 \quad 7/2 \quad 5/2 \quad \text{upper}$$

$$\frac{2}{7} (3/2)^{7/2} - \frac{4}{5} (3/2)^{5/2}$$

$$-\frac{117\sqrt{6}}{280}$$

$$u = 1 \quad 7/2 \quad 5/2 \quad \text{Lower}$$

$$\frac{2}{7} (1)^{7/2} - \frac{4}{5} (1)^{5/2}$$

$$-\frac{18}{35}$$



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$$30) \int_{\pi/2}^{3\pi/4} \sqrt{1 - \sin 2n} \, dn$$

Sol:-

$$1 - \sin 2n = (\sin n - \cos n)^2$$

$$\int_{\pi/2}^{3\pi/4} (\sin n - \cos n)^2 \, dn$$

$$\int_{\pi/2}^{3\pi/4} |\sin n - \cos n| \, dn$$

$$\int \sin n \, dn - \int \cos n \, dn$$

$$-\cos n + C - \sin n + C$$

$$-\cos n - \sin n + C \Big|_{\pi/2}^{3\pi/4}$$

$$\begin{array}{l} \text{Upper} \\ n = \frac{3\pi}{4} \\ -\cos\left(\frac{3\pi}{4}\right) - \sin\left(\frac{3\pi}{4}\right) \\ 0 \end{array}$$

$$\begin{array}{l} \text{Lower} \\ n = \frac{\pi}{2} \\ -\cos\left(\frac{\pi}{2}\right) - \sin\left(\frac{\pi}{2}\right) \\ -1 \end{array}$$

$$\begin{array}{l} \text{upper} - \text{Lower} \\ 0 - (-1) = 1 \end{array}$$



$$31) \int_0^{\pi/2} \theta \sqrt{1 - \cos 2\theta} d\theta$$

Sol:-

$$1 - \cos 2\theta = 2 \sin^2 \theta$$

$$\sqrt{2 \sin^2 \theta} \Rightarrow \sqrt{2} \sqrt{\sin^2 \theta} \Rightarrow \sqrt{2} \sin \theta$$

$$\int \theta \cdot \sqrt{2} \sin \theta d\theta \quad \sqrt{2} \int_0^{\pi/2} \theta \sin \theta d\theta$$

Integration by Parts

$$u = \theta \quad dv = \sin \theta d\theta$$

$$du = d\theta \quad , \quad v = -\cos \theta$$

$$\int_0^{\pi/2} \theta \sin \theta d\theta$$

$$\begin{aligned} \int u dv &= uv - \int v du \\ &= \theta(-\cos \theta) - \int (-\cos \theta) d\theta \\ &= -\theta \cos \theta + \int \cos \theta d\theta \\ &= -\theta \cos \theta + \sin \theta + C \end{aligned}$$

$$\begin{aligned} \theta = \pi/2 &= -\theta \cos \theta + \sin \theta + C \\ &= 1 \end{aligned}$$

$$\begin{aligned} \theta = 0 &= -\theta \cos \theta + \sin \theta + C \\ &= 0 \end{aligned}$$

$$\begin{aligned} \int_0^{\pi/2} \theta \sin \theta d\theta &\Rightarrow 1 - 0 = 1 \quad \checkmark \text{ upper - Lower} \\ \sqrt{2} \times 1 &= \sqrt{2} \end{aligned}$$

$$39) \int_{-\pi}^{\pi} (1 - \cos^2 t)^{3/2} dt$$

Sol:- $\int_{-\pi}^{\pi} (\sin^2 t)^{3/2} dt$ ~~$\Rightarrow \int_{-\pi}^{\pi} (\sqrt{\sin^2 t})^3 dt$~~ $\int_{-\pi}^{\pi} (\sin^2 t)^{3/2} dt$

$$\int_{-\pi}^{\pi} \sin^3 t dt$$

If function have odd power with the same limit but limit have different symbols like upper is positive lower is negative. So we break the limits.

$$\int_{-\pi}^0 \sin^3 t dt + \int_0^{\pi} \sin^3 t dt$$

If we put 0 or $-\pi$ between values into function then our answer will be in the subtraction

$$- \int_{-\pi}^0 \sin^3 t dt + \int_0^{\pi} \sin^3 t dt$$

$$- \int_{-\pi}^0 \sin^2 t \cdot \sin t dt + \int_0^{\pi} \sin^2 t \cdot \sin t dt$$

$$- \int_{-\pi}^0 (1 - \cos^2 t) \cdot \sin t dt + \int_0^{\pi} (1 - \cos^2 t) \sin t dt$$

$$- \int_{-\pi}^0 \sin t + \int_{-\pi}^0 \cos^2 t \sin t + \int_0^{\pi} \sin t - \int_0^{\pi} \cos^2 t \sin t$$



$$\cos t \int_{-\pi}^0 - \frac{\cos^3 t}{3} \int_{-\pi}^0 - \cos t \int_0^{\pi} + \frac{\cos^3 t}{3} \int_0^{\pi}$$

$$\left(1 - (-1) \right)^{\text{put Limits}} - \frac{1}{3} \left(1 - (-1) \right) - \left[(-1) - 1 + \frac{1}{3} (-1) - 1 \right]$$

$$\left[2 - \frac{1}{3} \cdot 2 + 2 - \frac{2}{3} \right]$$

$$2 - \frac{2}{3} + 2 - \frac{2}{3} \Rightarrow 4 - \frac{4}{3} \Rightarrow \frac{8}{3}$$